First-order gain model, $\lambda_f = 1$: Kalman filter, observability matrix, and the link

1 Model

Motion error (from 0921, δa_x removed)

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k], \qquad F_{dx}[k] = f_{dx}[k] + (1-\lambda_c)\sum_{i=1}^{d} f_{dx}[k-i]$$

Parameter model, estimator, error

$$a_x[k+1] = a_x[k], \qquad \hat{a}_x[k+1] = \hat{a}_x[k], \qquad e_{a_x}[k] = a_x[k] - \hat{a}_x[k], \qquad e_{a_x}[k+1] = e_{a_x}[k]$$

State equation

$$\begin{aligned}
\delta x_1[k+1] &= \delta x_2[k] \\
\delta x_2[k+1] &= \delta x_3[k] \\
\delta x_3[k+1] &= \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k] \\
a_x[k+1] &= a_x[k]
\end{aligned}$$

Output equation

$$\begin{aligned}
y_1[k] &= \delta x_m[k] = \delta x_1[k] + n_x[k] \\
y_2[k] &= a_{xm}[k] = a_x[k] + n_a[k]
\end{aligned}$$

Error model

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta x_1}[k] \\ e_{\delta x_2}[k] \\ e_{\delta x_3}[k] \\ e_{a_x}[k] \end{bmatrix}, \quad \mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{Q} = \mathrm{diag}\big(0,\ 0,\ Q_{33},\ 0\big), \quad Q_{33} = \mathrm{Var}(\varepsilon), \qquad \mathbf{R} = \mathrm{diag}\big(R_1,\ R_2\big), \quad R_1 = \mathrm{Var}(n_x),\ R_2 = \mathrm{Var}(n_a)$$

2 Kalman filter

Prediction

$$\begin{aligned}
\hat{\mathbf{x}}^f[k] &= \text{state equation applied to } \hat{\mathbf{x}}[k-1] \\
\mathbf{P}^f[k] &= \mathbf{F}_e[k-1]\,\mathbf{P}[k-1]\,\mathbf{F}_e^{\top}[k-1] + \mathbf{Q}
\end{aligned}$$

Update

$$\mathbf{L}[k] = \mathbf{P}^f[k]\,\mathbf{H}^\top\big(\mathbf{H}\,\mathbf{P}^f[k]\,\mathbf{H}^\top + \mathbf{R}\big)^{-1}$$
$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^f[k] + \mathbf{L}[k]\,\big(\mathbf{y}[k] - \mathbf{H}\,\hat{\mathbf{x}}^f[k]\big)$$
$$\mathbf{P}[k] = \big(\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}\big)\,\mathbf{P}^f[k]$$

3 The a_x element, y_2 only

Prediction of P_{aa}

$$\big(\mathbf{F}_e\,\mathbf{P}\,\mathbf{F}_e^\top\big)_{44} = [\,0\;\;0\;\;0\;\;1\,]\,\mathbf{P}\,[\,0\;\;0\;\;0\;\;1\,]^\top = P_{aa}$$
$$P^f_{aa}[k] = P_{aa}[k-1] + Q_{44} = P_{aa}[k-1]$$

Gain

$$\mathbf{H}_2\,\mathbf{P}^f\,\mathbf{H}_2^\top = P^f_{aa}$$
$$\ell_{42}[k] = \frac{P^f_{aa}[k]}{P^f_{aa}[k] + R_2}$$

Update of P_{aa}

$$P_{aa}[k] = \big(1 - \ell_{42}[k]\big)\,P^f_{aa}[k] = \frac{R_2}{P^f_{aa}[k] + R_2}\,P^f_{aa}[k]$$
$$\frac{1}{P_{aa}[k]} = \frac{P^f_{aa}[k] + R_2}{R_2\,P^f_{aa}[k]} = \frac{1}{P^f_{aa}[k]} + \frac{1}{R_2} = \frac{1}{P_{aa}[k-1]} + \frac{1}{R_2}$$
$$\frac{1}{P_{aa}[k]} = \frac{1}{P_{aa}[0]} + \frac{k}{R_2}$$

Gain written with $P_{aa}[k]$

$$\ell_{42}[k] = \frac{P^f_{aa}[k]}{P^f_{aa}[k] + R_2} = \frac{P_{aa}[k]}{R_2} = \frac{1/R_2}{\dfrac{1}{P_{aa}[0]} + \dfrac{k}{R_2}}$$

Update of $\hat{a}_x$

$$\hat{a}_x[k] = \hat{a}_x[k-1] + \ell_{42}[k]\,\big(y_2[k] - \hat{a}_x[k-1]\big)$$

Closed form (induction)

$$\text{claim:}\quad \hat{a}_x[k] = \frac{\dfrac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \dfrac{1}{R_2}\sum_{j=1}^{k} y_2[j]}{\dfrac{1}{P_{aa}[k]}}$$

$$\begin{aligned} k \to k+1: \quad \hat{a}_x[k+1] &= \big(1 - \ell_{42}[k+1]\big)\,\hat{a}_x[k] + \ell_{42}[k+1]\,y_2[k+1] \\ &= \frac{P_{aa}[k+1]}{P_{aa}[k]}\,\hat{a}_x[k] + \frac{P_{aa}[k+1]}{R_2}\,y_2[k+1] \\ &= P_{aa}[k+1]\left(\frac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \frac{1}{R_2}\sum_{j=1}^{k} y_2[j] + \frac{1}{R_2}\,y_2[k+1]\right) \end{aligned}$$

$$\left(1 - \ell_{42}[k+1] = \frac{R_2}{P_{aa}[k] + R_2} = \frac{P_{aa}[k+1]}{P_{aa}[k]}\right)$$

Substitute $y_2[j] = a_x[j] + n_a[j]$

$$\hat{a}_x[k] = \frac{\dfrac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \dfrac{1}{R_2}\sum_{j=1}^{k} a_x[j]}{\dfrac{1}{P_{aa}[0]} + \dfrac{k}{R_2}} \;+\; \frac{\dfrac{1}{R_2}\sum_{j=1}^{k} n_a[j]}{\dfrac{1}{P_{aa}[0]} + \dfrac{k}{R_2}}$$

4 Observability matrix

Definition (prediction only: $\mathbf{F}_e$, no L, no noise)

$$\boldsymbol{\Phi}(k_0 + j,\, k_0) = \mathbf{F}_e[k_0 + j - 1] \cdots \mathbf{F}_e[k_0], \qquad \boldsymbol{\Phi}(k_0, k_0) = \mathbf{I}, \qquad \mathbf{O}(k_0) = \begin{bmatrix} \mathbf{H}\,\boldsymbol{\Phi}(k_0, k_0) \\ \mathbf{H}\,\boldsymbol{\Phi}(k_0 + 1, k_0) \\ \mathbf{H}\,\boldsymbol{\Phi}(k_0 + 2, k_0) \\ \vdots \end{bmatrix}$$

The e_{a_x} column of $\boldsymbol{\Phi}$

$$\boldsymbol{\phi}[j] = \boldsymbol{\Phi}(k_0 + j, k_0)\,[\,0 \;\; 0 \;\; 0 \;\; 1\,]^\top = \begin{bmatrix}\phi_1[j] & \phi_2[j] & \phi_3[j] & \phi_4[j]\end{bmatrix}^\top$$

$$\boldsymbol{\phi}[0] = [\,0 \;\; 0 \;\; 0 \;\; 1\,]^\top$$

$$\boldsymbol{\phi}[j+1] = \mathbf{F}_e[k_0 + j]\,\boldsymbol{\phi}[j]$$

$$\phi_1[j+1] = \phi_2[j]$$

$$\phi_2[j+1] = \phi_3[j]$$

$$\phi_3[j+1] = \lambda_c \cdot \phi_3[j] - F_{dx}[k_0 + j] \cdot \phi_4[j]$$

$$\phi_4[j+1] = \phi_4[j]$$

$\phi_4[j] = 1$

$\phi_3[j] = -\sum_{i=0}^{j-1} \lambda_c^{j-1-i}\, F_{dx}[k_0+i]$

$\phi_2[j] = \phi_3[j-1]$

$\phi_1[j] = \phi_3[j-2]$

The e_{a_x} column of O

$\mathbf{H}_1\,\boldsymbol{\phi}[j] = \phi_1[j] = -\sum_{i=0}^{j-3} \lambda_c^{j-3-i}\, F_{dx}[k_0+i] \quad (j \geq 3), \qquad \mathbf{H}_1\,\boldsymbol{\phi}[0] = \mathbf{H}_1\,\boldsymbol{\phi}[1] = \mathbf{H}_1\,\boldsymbol{\phi}[2] = 0$

$\mathbf{H}_2\,\boldsymbol{\phi}[j] = \phi_4[j] = 1$

Constant F_{dx}

$$\phi_3[j] = -F_{dx}\,\frac{1-\lambda_c^j}{1-\lambda_c} \xrightarrow{\;j\to\infty\;} -\frac{F_{dx}}{1-\lambda_c}, \qquad \mathbf{H}_1\,\boldsymbol{\phi}[j] \;\to\; -\frac{F_{dx}}{1-\lambda_c}$$

F_{dx} at constant velocity (implementable control effort, $\delta x_m \approx 0$, $\hat{a}_x f_{dx}$ constant)

$$\hat{a}_x f_{dx} = \Delta x_d^d + (1-\lambda_c)\big[\,0 - d\,\hat{a}_x f_{dx}\,\big], \qquad \Delta x_d^d = \big(1+(1-\lambda_c)d\big)\Delta x_d$$

$$\hat{a}_x f_{dx}\,\big(1+(1-\lambda_c)d\big) = \big(1+(1-\lambda_c)d\big)\Delta x_d \;\;\Rightarrow\;\; f_{dx} = \frac{\Delta x_d}{\hat{a}_x}$$

$$F_{dx} = f_{dx} + (1-\lambda_c)\,d\,f_{dx} = \big(1+(1-\lambda_c)d\big)\,\frac{\Delta x_d}{\hat{a}_x}$$

5 Link: O inside the Kalman filter

y_2: the row of O is $\mathbf{H}_2$ itself ($\phi_4 = 1$)

$$\frac{1}{P_{aa}[k]} - \frac{1}{P_{aa}[k-1]} = \frac{1}{R_2} = \frac{\big(\mathbf{H}_2\,\boldsymbol{\phi}[j]\big)^2}{R_2}$$

y_1: $\mathbf{H}_1$ has 0 in the a_x column; the path is built in $\mathbf{P}^f$

$P_{3a}^f[k] = \big(\mathbf{F}_e\,\mathbf{P}\,\mathbf{F}_e^\top\big)_{34} = [\,0\;\;0\;\;\lambda_c\;\;-F_{dx}\,]\;\mathbf{P}\;[\,0\;\;0\;\;0\;\;1\,]^\top = \lambda_c\,P_{3a}[k-1] - F_{dx}[k-1]\,P_{aa}[k-1]$

$P_{2a}^f[k] = P_{3a}[k-1]$

$P_{1a}^f[k] = P_{2a}[k-1]$

prediction only, $P_{3a}[k_0] = 0$, P_{aa} constant over the window:

$P_{3a}[k_0+j] = \phi_3[j]\,P_{aa}, \qquad P_{1a}[k_0+j] = \phi_1[j]\,P_{aa} = \big(\mathbf{H}_1\,\boldsymbol{\phi}[j]\big)\,P_{aa}$

$$\ell_{41}[k] = \frac{P_{a1}^f[k]}{P_{11}^f[k]+R_1} \approx \frac{\big(\mathbf{H}_1\,\boldsymbol{\phi}[j]\big)\,P_{aa}}{P_{11}^f[k]+R_1}, \qquad \frac{1}{P_{aa}[k]} - \frac{1}{P_{aa}[k-1]} \approx \frac{\big(\mathbf{H}_1\,\boldsymbol{\phi}[j]\big)^2}{P_{11}^f[k]+R_1} \;\leq\; \frac{\big(\mathbf{H}_1\,\boldsymbol{\phi}[j]\big)^2}{R_1}$$

Both channels

$$\frac{1}{P_{aa}[k]} \approx \frac{1}{P_{aa}[0]} + \sum_{j=1}^{k} \big(w_2 + w_1[j]\big)$$

$$w_2 = \frac{1}{R_2}$$

$$w_1[j] = \frac{\big(\mathbf{H}_1\,\boldsymbol{\phi}[j]\big)^2}{P_{11}^f[j] + R_1} \approx \frac{1}{P_{11}^f + R_1}\left(\frac{1 + (1-\lambda_c)d}{1-\lambda_c}\,\frac{\Delta x_d[j]}{\hat{a}_x[j]}\right)^2$$

$$\hat{a}_x[k] \approx \frac{\frac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \sum_{j=1}^{k}\big(w_2 + w_1[j]\big)\,a_x[j]}{\frac{1}{P_{aa}[0]} + \sum_{j=1}^{k}\big(w_2 + w_1[j]\big)}$$

6 Speed

$$\Delta x_d = \frac{(h_0 - h_b)\,R\,T_s}{T_d}, \qquad \text{steps per } R \text{ travelled} = \frac{T_d}{(h_0 - h_b)\,T_s}$$

per R travelled: $\quad w_2\,\dfrac{T_d}{(h_0 - h_b)\,T_s} \;\propto\; T_d, \qquad w_1\,\dfrac{T_d}{(h_0 - h_b)\,T_s} \;\propto\; \Delta x_d^2\,T_d \;\propto\; \dfrac{1}{T_d}$

Figure: $\hat{a}_x$ vs running average of a_x

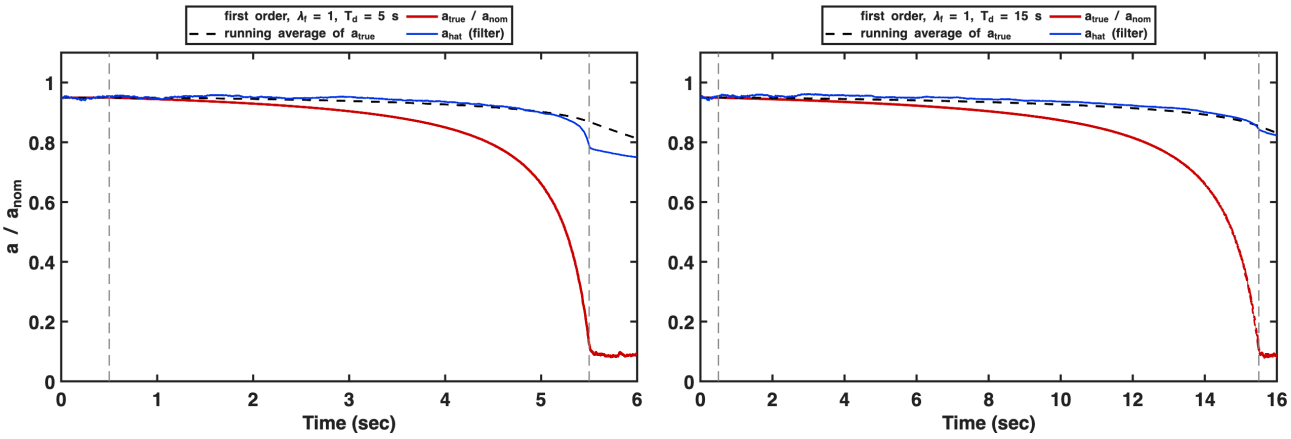


Figure: $1/P_{aa}$ added per step and per R travelled

